

Patient Profile

Saubhik Bhaumik, 27-year-old male

Summary of Results

Liver function tests reveal all measured parameters within reference ranges, indicating preserved hepatic synthetic and excretory function. ****ML Result: Normal**** – the predictive model assesses the profile as within normal limits.

Detailed Test Explanations

****Serum Bilirubin (Total)**:** Quantifies unconjugated and conjugated bilirubin, reflecting hepatic conjugation and excretion; elevations suggest hepatocellular injury, biliary obstruction, or hemolysis. Patient result: 0.9 mg/dl (reference: 0.2-1.2 mg/dl) – normal.

****Serum Bilirubin (Direct)**:** Measures conjugated bilirubin; increased levels indicate biliary obstruction or intrahepatic cholestasis. Patient result: 0.2 mg/dl (reference: 0-0.3 mg/dl) – normal.

****Serum Bilirubin (Indirect)**:** Represents unconjugated bilirubin; elevations occur in hemolysis or impaired conjugation (e.g., Gilbert syndrome). Patient result: 0.70 mg/dl (reference: 0.2-1.0 mg/dl) – normal.

****SGPT (ALT)**:** Alanine aminotransferase, a sensitive marker of hepatocellular injury; elevations indicate acute hepatitis or toxin exposure. Patient result: 36 U/L (reference: 13-40 U/L) – normal.

****SGOT (AST)**:** Aspartate aminotransferase, found in liver and cardiac muscle; rises with hepatocellular damage, though less specific than ALT. Patient result: 32 U/L (reference: 0-37 U/L) – normal.

****Serum Alkaline Phosphatase**:** Enzyme elevated in biliary obstruction, bone disease, or pregnancy; reflects cholestasis or infiltrative liver processes. Patient result: Not available.

****Serum Protein**:** Total protein level assesses nutritional status and hepatic synthetic function; hypoalbuminemia suggests chronic liver disease. Patient result: 7.2 g/dl (reference: 6.4-8.3 g/dl) – normal.

****Serum Albumin**:** Primary protein synthesized by hepatocytes; low levels indicate chronic liver disease, malnutrition, or nephrotic syndrome. Patient result: 4.7 g/dl (reference: 3.5-5.2 g/dl) – normal.

****Globulin**:** Immunoglobulin fraction; elevations occur in chronic infections or autoimmune liver disease. Patient result: 2.50 g/dl (reference: 1.8-3.6 g/dl) – normal.

****A/G Ratio**:** Albumin-to-globulin ratio; reversal suggests chronic liver disease or inflammation. Patient result: 1.88 (reference: 1.1-2.4) – normal.

Overall Interpretation

Comprehensive liver function profile demonstrates normal bilirubin metabolism, transaminase activity, and protein synthesis, consistent with intact hepatic function. No evidence of hepatitis, cholestasis, or synthetic dysfunction. ****ML Result: Normal**** reinforces absence of abnormal patterns requiring immediate concern; correlate with clinical history.

Recommendations

Maintain balanced diet rich in fruits, vegetables, and lean proteins; limit alcohol intake to <14 units weekly; engage in regular aerobic exercise (150 minutes/week); monitor weight to prevent fatty liver. Routine follow-up liver function testing advised annually or if symptoms (jaundice, fatigue) arise. Consult physician for personalized correlation.

Disclaimer

This report provides interpretive guidance based on provided data and is not a substitute for professional medical advice, diagnosis, or treatment. Always seek evaluation by a qualified healthcare provider.